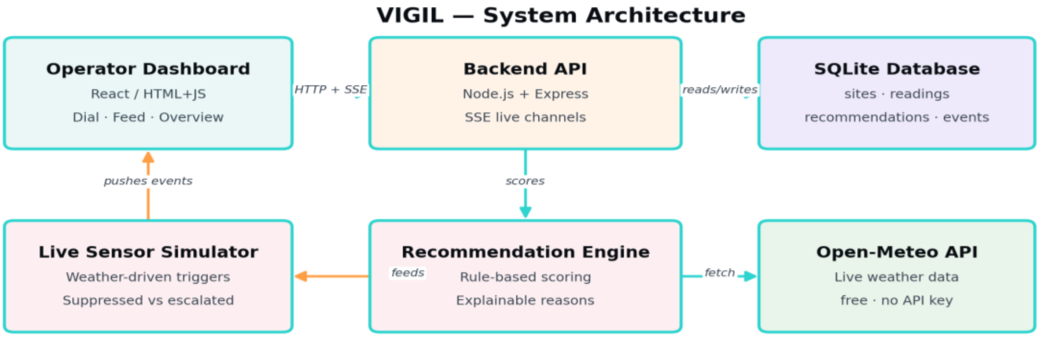
A-1 LAUNCHPAD 2026

National Hackathon — Round 2 Case Study Submission

VIGIL

Weather-Based Sensor Calibration Suggestion System



Track: Software Development — AI / Machine Learning

Team Name: **Team Unstoppable**

College / Institution: Sanjivani College Of Engineering, Kopargaon

Team Members: 1)Kunal Sharad Gawand

2)Samruddhi Bharat Kokate

VIGIL — Weather-Based Sensor Calibration Suggestion System

A live, self-adjusting calibration layer for Perimeter Intrusion Detection Systems (PIDS)

1. Problem Statement

Perimeter Intrusion Detection Systems (PIR / microwave / dual-tech sensors) generate a large share of their false alarms from weather: wind-blown debris, heavy rain sheeting on lenses, storm gusts, and fog. Operators today adjust sensor sensitivity manually and reactively — after alarm fatigue has already set in. There is no live, explainable, weather-aware layer between the forecast and the sensor configuration.

2. Our Solution

VIGIL is a live calibration module that continuously reads real weather conditions for each monitored site, runs them through a transparent rule engine, and recommends — or automatically applies — the sensor sensitivity best suited to current conditions. Every recommendation ships with plain-language reasoning, not a black-box score. A simulated live sensor/incident feed (driven by real weather data) demonstrates the suppressed-vs-escalated decision in real time, and a 24-hour forecast lookahead lets operators see a sensitivity change coming before it happens.

3. System Architecture

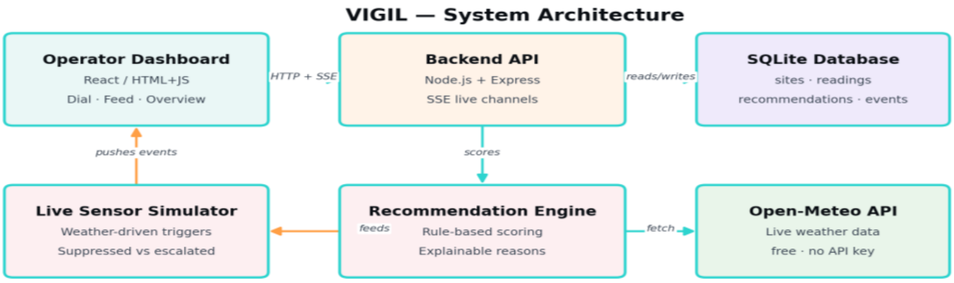


Fig. 1 — VIGIL system architecture: dashboard, API, rule engine, database, live sensor simulator, and the external weather source.

4. What Makes This Dynamic, Not Static

- Server-Sent Events push weather updates, sensitivity changes, and simulated sensor events to every connected dashboard instantly — no manual refresh.
- A live activity feed shows individual zone-level sensor triggers, reasoned in real time against the current sensitivity level (suppressed vs. escalated for operator review).
- A mission-control overview strip shows all monitored sites at a glance with live status dots, updating over its own SSE channel.
- A 24-hour forecast lookahead predicts the next likely sensitivity change (e.g. “rain expected in 3h — sensitivity will auto-lower to Medium”), giving operators advance notice.
- Manual override is a first-class, auditable action — operators can always take control, and every override is logged with a timestamp and reason.

5. Live Dashboard

Fig. 2 — VIGIL dashboard running live: multi-site overview strip, live weather conditions, sensitivity dial with reasoning, 24h predictive forecast alert, and trend + analytics — all populated with real data from Open-Meteo.

6. Recommendation Engine

Sensitivity starts at High and is stepped down as adverse conditions accumulate. Every rule that fires is surfaced to the operator in plain language, so the recommendation is explainable rather than a black box.

Condition	Threshold	Effect on sensitivity
Wind speed	≥ 20 km/h (moderate) / ≥ 40 km/h (severe)	-1 / -2
Wind gusts	≥ 55 km/h	-1
Precipitation	≥ 2 mm (light) / ≥ 15 mm (heavy)	-1 / -2
Thunderstorm	Weather code ≥ 95	-2, storm flag raised
Humidity + rain	≥ 90% while raining	-1 (lens fogging risk)
Temperature	≤ 2°C	-1 (frost/ice risk)

7. Tech Stack

Layer	Choice	Why
Frontend	HTML/JS + Chart.js (React variant also included)	Fast, dependency-light, real-time-friendly
Backend	Node.js + Express	Simple, well-understood REST + SSE layer
Database	SQLite (better-sqlite3)	Zero-config, file-based, ideal for a judged prototype
Weather data	Open-Meteo API	Free, no API key, CORS-enabled — reliable for live demos
Real-time transport	Server-Sent Events	Simple one-directional push, native browser support, auto-reconnect
Scheduling	node-cron	Configurable polling independent of dashboard being open

8. Live Activity Feed in Action

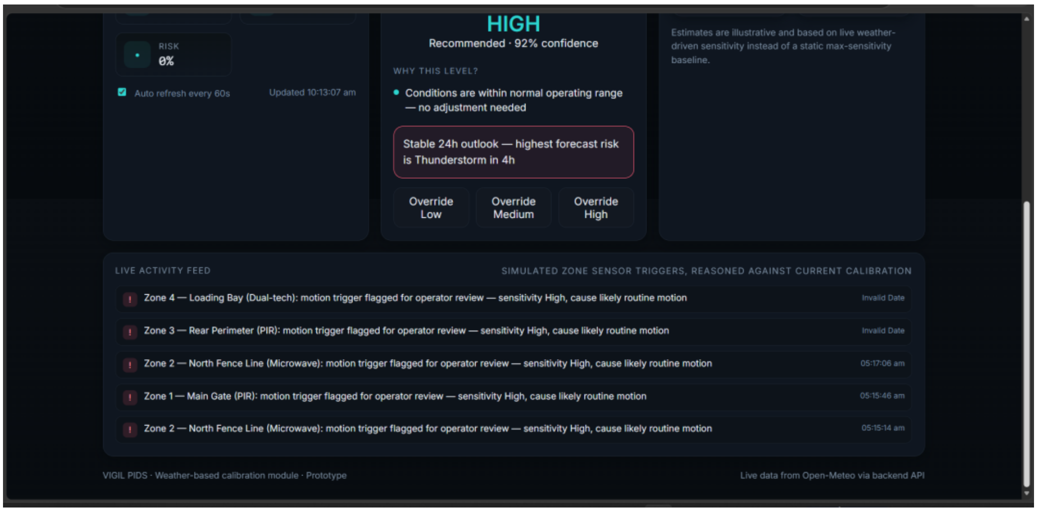


Fig. 3 — Live simulated zone-sensor triggers streaming into the dashboard in real time via Server-Sent Events, reasoned against the current sensitivity level for each of the 4 perimeter zones.

9. Impact & Analytics

- Estimated false-alarm reduction is modeled from the proportion of readings where live conditions triggered a lowered sensitivity, relative to a static always-High baseline — presented to judges as an illustrative model, not a measured result, pending real operator log data.
- Every automatic sensitivity change and storm detection is logged as an auditable event, alongside manual overrides, giving a full timeline for post-incident review.
- The live sensor/incident feed is explicitly a simulation calibrated to real weather conditions (no physical sensors are wired into this prototype) — it demonstrates the suppressed-vs-escalated decision the calibration engine would drive in a live deployment.

10. What's Next

- Replace/augment the rule engine with a model trained on real historical false-alarm logs once available from a pilot site.
- Direct integration with camera/sensor firmware to apply sensitivity changes automatically rather than via operator confirmation.
- Multi-tenant authentication, SMS/email alerting, and role-based access for security operations centers.

11. Third-Party Acknowledgements

Weather data: Open-Meteo.com (free API, CC BY 4.0 attribution). Charting: Chart.js / Recharts. Icons: Lucide.

12. Links

GitHub Repository: https://github.com/Kunalgawand03/Weather-Based-Sensor-Calibration-Suggestion-System.git

Video Demonstration:
https://drive.google.com/file/d/1FSiVfLJlvB8oCh4fvoc1r0Zbh5LQVgRf/view?usp=drivesdk

Kunal Sharad Gawand

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Professional Summary

Results-driven Electronics and Computer Engineering student (CGPA: **8.48/10**) with Honors in Cyber Security, specializing in full-stack web development using the MERN stack and AI/ML integration. Proficient in application design, front-end & back-end development, debugging, REST API development, and deployment.

Education

Sanjivani College of Engineering, Kopargaon — July 2023 – July 2027
B.Tech in Electronics and Computer Engineering (Honors: Cyber Security) — *CGPA: 8.53 / 10.0*
Shree Chhatrapati Shivaji Junior College, Kolpewadi — 2023
Higher Secondary Certificate (Science) — *Maharashtra State Board*

Technical Skills

Languages: Java, JavaScript (ES6+), Python, C, CSS3, HTML5
Web & Backend: React.js, Node.js, Express.js, FastAPI, REST APIs
Databases: MySQL, MongoDB, PostgreSQL
DevOps & Cloud: AWS, Git, GitHub, VS Code, Android Studio,
AI/ML & IoT: Python (OpenCV, YOLOv8), OpenAI GPT, RAG Pipelines, IoT Sensors
DSA: Arrays, Linked Lists, Queues and Stacks, Trees, Strings, Graphs, Sorting Algorithms

Projects

ShikshakMitra AI | *Python, FastAPI, React.js, YOLOv8, OpenCV, GPT, RAG, PostgreSQL, Redis, Docker, AWS* — Dec 2025

- Engineered a **multi-modal AI system** leveraging YOLOv8, OpenCV, speech recognition, and NLP to perform real-time classroom engagement analysis and evaluate teaching effectiveness.
- Designed and implemented a **GPT-based RAG pipeline** to auto-generate personalized teaching insights, performance analytics, and bias-aware evaluation reports for educators.
- Architected a **scalable microservices platform** with FastAPI, React.js, Plotly, Docker, Redis, and AWS, delivering live analytics through an interactive dashboard with sub-second response times.
- **National Recognition:** Finalist at IIT Bombay Techfest (Asia's largest tech festival) for this project, competing among top engineering teams nationwide.

Airbnb – Full-Stack Web Application | *JavaScript, Node.js, Express.js, MongoDB, RESTful APIs, Git* — Aug 2025

- Developed a **full-stack property rental platform** with secure JWT-based authentication, dynamic property listings, search/filter functionality, and end-to-end booking workflows.
- Built and consumed **RESTful APIs** enabling seamless client-server communication with efficient data handling across devices; implemented MVC architecture for clean, maintainable code.
- Delivered a **fully responsive UI** using JavaScript and CSS, ensuring smooth cross-device user experience simulating a production-grade platform.

Achievements & Hackathons

National Finalist – AWS Impact X Challenge — IIT Bombay Techfest
Built an AI-based real-time safety & risk prediction system; recognized among top teams nationally.
Top 5 Finalist – IIT Bombay Techfest (ShikshakMitra AI Track) — IIT Bombay
*Led a **Top 5** team building an AI-based teaching assistant improving teacher productivity and classroom efficiency.*
Finalist – Hackatron, Infotsav'25 (IIITM Gwalior) — Oct 2025
Selected in Top 100 teams out of 800+ teams (6,000+ participants) in a national GitHub-powered hackathon.

Certifications

Oracle Cloud Infrastructure 2025 AI Foundations Associate — Oct 2025
Oracle University – Cloud computing, AI fundamentals, and OCI infrastructure
Honeywell Future Skills Edge Program – SE Job Simulation — Apr 2026
Honeywell – Industry-aligned software engineering tasks and professional development
Web Fundamentals — Dec 2025
TryHackMe – Hands-on web development and HTTP fundamentals via cybersecurity platform
Project Management — Jan 2026
Foundations of project planning, execution, monitoring, and delivery

Extracurricular Activities

ECESA – Social Media Manager (July 2025 – Present): Content creation, digital promotions, and community engagement for Engineering department.
SAC – Organizing Committee Member (July 2025 – Present): Event coordination and logistics for college-level academic and cultural activities.

Samruddhi Kokate

Mechanical Design Engineer – CAD & Product Design

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Career Objective

Motivated Mechanical Engineering graduate (B.Tech, 2026) with hands-on experience in CAD design (SolidWorks, Creo Parametric), Special Purpose Machine (SPM) design, IoT-based systems, and a filed/granted design patent. Seeking an entry-level Mechanical Design Engineer role to apply machine design, GD&T, and problem-solving skills to deliver measurable value in product design and manufacturing.

Education

Institution / Degree	Year	Score
Sanjivani College of Engineering, Kopargaon – B.Tech in Mechanical Engineering	2026	7.67/10
Radhabai Junior College, Kolpewadi – HSC (Maharashtra State Board)	2023	58.67%
New English School, Karwadi – SSC (Maharashtra State Board)	2021	85.20%

Professional Training

Machine Tool Design (SPM) – IMTMA, Pune *Ongoing*

- Hands-on training in design and manufacturing of Special Purpose Machines (SPMs)
- Spindle design and selection of bearings, ball screws, guideways, and couplings
- Manufacturing and assembly drawings, BOM preparation, and GD&T
- Fundamentals of FEA, hydraulics, pneumatics, and electrical systems
- Live project: Design of an SPM Milling Machine

Patent

Wheat Chaff Remover Machine – Design Patent Filed / Granted

- Designed an agricultural machine that separates chaff from wheat with minimal grain loss using airflow and mechanical separation techniques
- Improved wheat quality and increased yield, enabling scalable farming operations

Projects

IoT-Based WiFi Toy Car *Aug 2024 – Nov 2024*

- Built a mobile-app-controlled toy car using the ESP8266 WiFi module for real-time wireless control
- Designed and implemented motor driver circuitry to enable smooth directional movement
- Integrated IoT communication protocols to link mobile app commands with hardware response
- *Tools/Tech:* IoT, ESP8266, Mobile App Control

Battery Health & Monitoring System *June 2025 – Nov 2025*

- Developed a real-time monitoring system tracking voltage, current, and temperature for battery health analysis
- Incorporated visual status indicators to enhance safety, performance, and battery lifespan
- Utilized sensors and controller modules for continuous data tracking and early fault detection

Skills

CAD & Design Tools: SolidWorks, Creo Parametric, Machine Tool Design
Programming: Python
Core Competencies: GD&T, Problem-Solving, Teamwork

Certifications

- CAD Master Participation Certificate – K.K. Wagh College of Engineering — June 2025
- NPTEL Certificate in Project Management – IIT Kanpur — Sep 2025
- Creo Parametric (40 hrs) – Sanjivani College of Engineering — Feb 2025
- Web Fundamentals (20 hrs) – TryHackMe Learning Path — Dec 2025
- Ethical Hacking Essentials – Coursera
- Artificial Intelligence Fundamentals — Feb 2026

Languages

English • Marathi • Hindi • Japanese (N5)